

YUVRAJ BANSAL

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Technical Skills

- **Languages & Core CS:** C/C++, Python, JavaScript, TypeScript — Data Structures & Algorithms, OOPs
- **Full-Stack Development:** React.js, Next.js, Node.js, Express.js, Tailwind CSS, WebSockets, REST APIs, tRPC
- **Databases & Cloud/DevOps:** PostgreSQL, MongoDB, Redis, Prisma — Kafka, Docker, Kubernetes, Terraform, Git
- **Generative AI & ML:** LLM / Generative AI SDK, LangChain, LangGraph, Vector DBs, RAG

Experience

Equivalic

Mar, 2026 – Present

Full Stack Developer Intern

Silicon Valley, United States (Remote)

- Engineered a canvas Architecture Studio with viewport and "Fit-to-Screen" algorithms to map legacy transformations.
- Optimized client-side parsing of deep JSON payloads, **reducing TTI by 42%** under heavy data loads.
- Architected a global appState engine to sync tabs, UI, and breadcrumbs, **reducing technical debt by 30%**.

Judix / IIT Patna Incubation

Jan, 2026 – Feb, 2026

Full Stack Developer Intern

Delhi, India (Remote)

- Architected Next.js full-stack authentication with a multi-channel (WhatsApp/SMS/Email) OTP backend.
- Enforced strict client/server schema validation via Zod, **eliminating 100% of invalid database data payloads**.
- Designed custom API middleware for secure session handling, **slashing authentication API latency by 25%**.

Valsco Technology Pvt Ltd

Jun, 2025 – Aug, 2025

Frontend Developer Intern

Chennai, India (Remote)

- Deployed a 3D Virtual Try-On feature for lenses, **enhancing web rendering and asset load performance by a measurable 20%**.
- Integrated frontend interfaces with backend REST APIs using Async/Await, **boosting cross-browser responsiveness by 15%**.

Projects

Distributed Event-Driven Automation Engine — Node.js, trpc, PostgreSQL, Drizzle, Kafka, Docker, K8s, ArgoCD

- Architected a **fault-tolerant microservices platform** processing real-time workflows using the **Transactional Outbox Pattern** across PostgreSQL and **Apache Kafka** to eliminate event loss.
- Engineered high-throughput ingestion services featuring **Redis rate limiting for webhook** defense and custom OAuth state locks for **Automated API polling triggers**.
- Integrated stateless worker engines leveraging an **Generative AI SDK** for automated payload summarization dispatched via Slack, Discord, and SMTP webhooks.
- Deployed cloud-native infrastructure using **Terraform IaC, ArgoCD GitOps, and Argo Rollouts** for zero-downtime Canary deployments, monitored via **Prometheus & Grafana**.

Multi-Tenant Feedback SaaS — Next.js, Node.js, PostgreSQL, Redis, Socket.io, BullMQ, OpenFGA



- Architected a **Multi-Tenant B2B SaaS** with shared-schema **Postgres RLS** via Prisma Extensions, eliminating **100%** of invalid cross-tenant queries at the ORM layer without developer error.
- Engineered granular ReBAC with OpenFGA (Google Zanzibar) + Clerk and a **horizontally scalable** real-time engine via **Socket.io + Redis Pub/Sub**, replacing SQL ACL joins with sub-millisecond cached permission traversals.
- Implemented a Redis-backed **BullMQ Job Queue** with dedicated **Background Workers** for outbound webhooks and audit logs, dropping API response times from **10s to under 5ms** by offloading tasks off the Node.js event loop with exponential backoff retries guaranteeing delivery.
- Designed a cryptographic API key system with Redis INCR usage tracking enforcing billing limits **10× faster** than Postgres queries, and **Razorpay webhook-driven** billing to auto-fulfill subscriptions server-to-server.

Achievements

Semi-Finalist, RIFT Hackathon 2026 — Physics Wallah

Feb, 2026

- Built a blockchain-based attendance system using Algorand with secure wallet integration.

Winner, Best Fresher Team, Reverse Coding Competition — Yantra '25, VIT Vellore

Feb, 2025

- Developed optimized solutions by deriving logic from input-output constraints.

Winner, Project 2039 Hackathon — Gravitas, VIT Vellore

Nov, 2024

- Built an IoT-based environmental monitoring solution leveraging CRISPR-driven microplastic detection.

Education

Vellore Institute of Technology

Aug 2024 – Oct 2028

B.Tech in Computer Science and Engineering

Vellore, India

Extracurricular / Leadership

Web Development Lead

Aug, 2025 – Present

Mozilla Firefox Club

Vellore, India

- Increased club engagement by 20% by organizing technical events like Playnatics and CodeXcape.
- Led maintenance of club web platforms, ensuring responsive UI frameworks and reliable performance.